CPROGRAM QUIZ – 1

Question 1 How many numbers can print after a point using %.2f?

- 2
- 3
- 4
- 1

Question 2 What will be the output of the following code if we provide input value **10 for a** and **20 value for b**?

```
#include<stdio.h>
int main()
{
   int i,a,b,j,k;
   i=printf("\n%d",printf("SunbeamInfo"));
   j = printf("\n%d ", i);
   k= printf("\n%d ", scanf("%d%d",&a,&b));
   printf("%d %d",j,k);
   return 0;
}
```

- SunbeamInfo 11 3 2 3 3
- 11 3 2 3 3
- Compile time Error
- Runtime Error

Question 3 What will be the output of the following code?

```
#include<stdio.h>
int main( )
{
   unsigned int num=0;
   int i=1;
   printf ("%d", (num >> i & 1 << 15 ) ? 1 : 0);
}
```

- It prints all even bits from num
- It prints all odd bits from num
- It prints binary equivalent of num
- prints 0

Question 4 What will be the output of the following code?

```
#include<stdio.h>
int main(){
   int n=1;
   printf("%d %d %d %d",n,n*n,n*=n,n++);
   return 0;
}
```

- 2 4 4 1
- 16 16 4 2
- 4 16 4 1
- 4 16 4 2

Question 5 Which of the following can be considered as a derived data type?

- array
- short int
- enum
- signed int

Question 6 The Data types describe about variable is:

- Type of data
- Types of operation
- Memory require
- All of the above

Question 7 The type modifier we can use with double?

- unsigned
- long
- short
- All of the above

Question 8 Pre increment / Post decrement operator are:

- Arithmetic Operator
- Misc Operator
- Relational Operator
- Bitwise Operator

Question 9 The scanf() function is declared in _____ header file.

- stdlib.h
- ctype.h
- stdio.h
- stdarg.h

Question 10 The printf() function returns:

- Number Of characters printed to the standard output.
- Number Of characters excluding the digits, spaces and punctuation.
- Number of spaces.
- None of these

CPROGRAM QUIZ – 2

Question 1: In switch case the case values can be:

- Integer
- Float
- Character
- Both 1 & 3

Question 2: What will be the output of following program?

```
#include<stdio.h>
enum menu
{ a=2,b,c,d=-1,e};
enum menu m;
int main()
{
  printf("%d %d",b+a,e+d);
}
```

- 2 3
- 0 1
- 5 -1
- Error

Question 3: What will be the output of following program?

```
#include<stdio.h>
int main()
{
  char ch;
  if((~sizeof(ch) + !sizeof('A')))
    printf("Hinjewadi");
  else
    printf("MarketYard");
  return 0;
}
```

- MarketYard
- Hinjewadi
- NO OUTPUT
- HinjewadiMarketYard

Question 4: What will be the output of following program?

```
#include <stdio.h>
void main()
{
  int a=10;
  switch(a){
    case 5+5:
      printf("Pune\n");
    default:
      printf("Karad\n");
  }
}
```

- PuneKarad
- Pune
- Karad
- Compile time error.

Question 5: What will be the output of following program?

```
#include <stdio.h>
void main()
{
  !printf("") ? printf("Sunbeam") : printf("Karad");
}
```

- Sunbeam
- karad
- 1
- 0

Question 6: What will be the output of the following C code?

```
#include<stdio.h>
enum test
{
  a=2,b=3.56
};
enum test t;
main()
{
  printf("%d%d",a,b);
}
```

- 2 3
- 0 1
- 2 3.56
- Error

Question 7: Find the correct statement from below:

- **I.** Decision control helps to compile code based on expression check at runtime.
- **II.** Decision control helps to execute code and compile time based on expression check.
- **III.** Decision control compiles the entire code no matter whether the expression check is satisfied or not.
- **IV.** We can not use the break jump statement with switch decision control.

Question 8: Enumeration is a _____ data type in C.

- User defined
- Built-in
- Library
- None Of the above

Question 9: What will be the output of following program?

```
#include <stdio.h>
int main() {
 int x = 15;
 if (x < 10)
  printf("kdac");
 else if (x >= 10 && x < 20)
  printf("dbda");
 else if (x >= 20 && x < 30)
  printf("daC");
 else
  printf("predac");
 return 0;
}
```

- kdac
- dbda
- daC
- predac

Question 10: What will be the output of following program?

```
#include <stdio.h>
int main()
{
 int a = 0;
 switch (++a, a)
 {
  case 0:
   printf("pune\n");
   break;
  case 1:
   printf("marketYard\n");
  default:
   printf("karad\n");
   break;
 }
 return 0;
}
```

- pune
- marketYard
- karad
- marketYard (followed by) karad

CPROGRAM QUIZ – 3

1. What is the output of the following code?

```
#include<stdio.h>
void main()
{
   int s=0;
   while(s++<10)
   {
      if(s<4 && s<9)
         continue;
      printf("\n%d\t",s);
   }
}
```

2. Point out the error, if any, in the while loop.

```
main()
{
   int i=1;
   while()
   {
      printf("%d",i++);
      if ( i > 10 )
         break;
   }
}
```

- There should be at least a semicolon in the while().
- The while loop should be replaced by a for loop.
- No error.
- Compile Time Error

3. What will be the output of the following code?

```
#include <stdio.h>
int main() {
  int count = 0, a = 0, b = 1;
  for (;;) {
    printf("#");
    if (count == a % 2 ? a++ : a-- ? a = 0 : ++b ? b = 2 : b++)
      break;
    printf("%d", ++count >> 1);
  }
  return 0;
}
```

- #0#1#2#3#4#5#
- #0#1#2#3#4#5#6###
- #0#1#2#3#4#5#6#7#

4. What will be the output of following code?

```
int main() {
  int x;
  for (x = 1; x <= 5; x++) {
    switch (x) {
      case 1:
        printf("Sunbeam");
        continue;
      case 2:
      case 3:
      case 4:
        break;
      case 5:
        printf("Info");
    }
    printf(".com");
  }
  return 0;
}
```

- SunbeamInfo.com
- https://www.google.com/search?q=Sunbeam.com.com.comInfo.com
- https://www.google.com/search?q=Sunbeam.comInfo.com
- https://www.google.com/search?q=Sunbeam.com.com.com.comInfo.com

5. Which of the following way is correct to make use of do ...while loop?

- int main() { do { printf("Inside loop"); }while(1) }
- int main() { do { printf("Inside loop"); while(1) } }
- int main() { do { printf("Inside loop"); }while(1); }
- int main() { { printf("Inside loop"); }while(1); }

6. What will be the output of the following?

```
#include<stdio.h>
int main()
{
   int i=48;
   for(printf("Sunbeam "); i <= 57 ; i+=3 )
   {
      printf("%c ",i);
   }
   return 0;
}
```

- 0 1 2 3 4 5 6 7 8 9
- Sunbeam
- Sunbeam 0 3 6 9
- Compile Time Error

7. What will be the output of the following:

```
#include<stdio.h>
int main()
{
   int a = 9;
   do
   {
      printf("%d ", a);
      a--;
   }while(a >= 5);
   return 0;
}
```

- 9
- 9 8 7 6 5
- No Output
- Compile Time Error

8. Which of the following are jump statements?

1. return
2. break
3. NULL
4. goto

- 4 only
- 2 and 3
- 1, 2, 4
- All of the above

9. What is the correct way to write a while loop that prints the numbers 1 to 10?

- int i = 1; while (i <= 10) { printf("%d ", i); i++; }
- int i = 1; while (i < 10) { printf("%d ", i); i++; }
- int i = 1; while (i >= 10) { printf("%d ", i); i++; }
- int i = 1; while (i == 10) { printf("%d ", i); i++; }

10. What will be the output of the following?

```
#include <stdio.h>
int main() {
   int count = 0;
   for (int i = 1; i <= 10; i++) {
      if (i % 3 == 0) {
         count++;
         continue;
      }
   }
   printf("Count: %d\n", count);
   return 0;
}
```

- Count: 2
- Count: 3
- Count: 4
- Count: 5

CPROGRAM QUIZ – 4

Question 1: What is the output of the following code?

```
#include <stdio.h>
int main()
{
   int x = 10;
   static int y = x;

   if(x == y)
      printf("Equal");
   else if(x > y)
      printf("Greater");
   else
      printf("Less");
   return 0;
}
```

- Compiler Error
- Equal
- Greater
- Less

Question 2: What is the output of the following code?

```
#include <stdio.h>
int a, b, c = 0;
void prtFun (void);
int main ()
{
   static int a = 1; /* line 1 */
   prtFun();
   a += 1;
   prtFun();
   printf ( "n %d %d " , a, b) ;
}

void prtFun (void)
{
   static int a = 2; /* line 2 */
   int b = 1;
   a += ++b;
   printf (" n %d %d " , a, b);
}
```

- 3 14 14 2
- 4 26 16 1
- 4 26 2 2 0
- 3 15 25 2

Question 3: What will it print?

```
#include <stdio.h>
int doSomething(int a, int b)
{
   if (b==1)
      return a;
   else
      return a + doSomething(a,b-1);
}

int main()
{
   int a = doSomething(2,3);
   printf("%d",a);
}
```

- 2
- 8
- 6
- 0

Question 4: What is the output of the following code?

```
#include <stdio.h>
void myshow(int);

int main()
{
   int a=10;
   printf("%d ", a);
   myshow(a);
   printf("%d", a);
}

void myshow(int k)
{
   k=20;
}
```

- 20 10
- 20 20
- 10 20
- 10 10

Question 5: In general, in a recursive and non-recursive implementation of a problem (program):

- Both time and space complexities are better in recursive than in non-recursive program.
- Both time and space complexities are better in non-recursive than in recursive program.
- Time complexity is better in recursive version but space complexity is better in non-recursive version of the program.
- Space complexity is better in recursive version but time complexity is better in non-recursive version of the program.

Question 6: What will be the output of this code?

```
#include <stdio.h>
int main()
{
   int i,x=10;
   i=abc(++x);
   printf("%d",--i);
   return 0;
}

int abc(int i)
{
   return(i++);
}
```

- 10
- 9
- 11
- none of the above

Question 7: What is the output?

```
#include<stdio.h>
int main(){
   static char i;
   while(i<=10)
      (i>2)? i++:i--;
   printf("%d ",i);
}
```

- 10
- 127
- 2
- 0

Question 8: What is the purpose of the extern keyword in C?

- To declare a variable or function that is defined in another file or scope.
- To allocate memory for a variable.
- To make a variable local to a function.
- To initialize a variable to zero.

Question 9: What is the output?

```
#include <stdio.h>
int main()
{
   int x = 22;
   register int static y = 20 ;
   printf("y = %d", ++y);
   return 0;
}
```

- Y = 20
- Error : multiple storage classes are not allowed
- Y = 21
- Error : Register cannot be used with int.

Question 10: What happens when a variable is passed by address to a function in C?

- A copy of the variable is made and passed to the function
- The memory address of the variable is passed to the function
- The value of the variable is passed to the function
- The variable is deleted and recreated in the function

CPROGRAM QUIZ – 5

Question 1 What will be the O/P of the following code?

```
#include<stdio.h>
int main()
{
   int arr[8] = {1,2,3,4,5,6,7,8};
   printf("%d",(arr+1)[3]);
}
```

- 3
- 4
- 5
- Compile Error

Question 2 An array elements are always stored in _____ memory locations.

- Sequential
- Random
- Both A and B
- None of the above

Question 3 What will be the output of the following code?

```
#include<stdio.h>
int main()
{
   int i;
   int arr[5]={1,2,3,4,5};
   *arr+=*(arr+3)-2[arr];
   for (i = 0; i < 5; ++i)
   {
      printf("\t%d",arr[i]);
   }
}
```

- 2,2,3,4,5
- 1,2,3,4,5
- Compile time error
- 3,3,3,4,5

Question 4 What will be the output?

```
#include <stdio.h>
int main(void)
{
   int i=0;
   char *name = "Sunbeam";
   name[i]='A';
   printf("%s",name);
}
```

- Compile time Error
- Sunbeam
- Aunbeam
- Runtime Error

Question 5 What will be the output?

```
#include <stdio.h>
int main(void)
{
   char str1[]="Sunbeam";
   printf("%d",*(str1+strlen(str1)));
}
```

- Garbage
- 8
- 7
- 0

Question 6 What will be the output?

```
#include <stdio.h>
int main(void)
{
   char name[5];
   int i;
   char j=65;
   for (i = 0; i < 4; ++i)
   {
      name[i]=j;
      j++;
   }
   i=0;
   name[i]='\0';
   printf("%s",name);
}
```

- ABCD
- A
- ABC
- No Output

Question 7 What will be the output of this code?

```
int main(void)
{
   double arr[4]={11.1, 22.2, 33.3, 44.4};
   double *p = arr+2;
   p -= 1;
   *p++ = 30.30;
   printf("%.2f %.2f",*p,arr[2]);
   return 0;
}
```

- 30.30 22.20
- 30.30 33.30
- 33.30 33.30
- 22.20 30.30

Question 8 Find the output of the following:

```
int main()
{
   int arr[]={11,22,33,44,55};
   int *ptr;
   printf("%d %d %d",2[arr],1[ptr],*(ptr=arr+2));
}
```

- 33 44 33
- 44 44 44
- 33 33 33
- Garbage garbage garbag

Question 9 What will be the output?

```
#include<stdio.h>
#include<string.h>
int main()
{
   printf("%c","SUNBEAM"[2]);
}
```

- U
- N
- SUNBEAM
- Complier Error

Question 10 Which is the correct way to [initialize] string at the time of declaration? I. char str1[]="DBDA"; II. char str3[4]= {'D','B','D','A'}; III. char str2[10]="DBDA"; IV. char str4[]= {'D','B','D','A'};

- only I, II and III
- only I, II and IV
- only I, II, III, IV
- only I, III

CPROGRAM QUIZ – 6

Question 1

```
#include <stdio.h>
int main(void)
{
   int num=10;
   int *ptr=&num;
   ++*ptr++;
   printf("%d",num);
   printf("%d",*--ptr);
}
```

- A) lvalue required error
- B) 10 garbage
- C) 11 12
- D) 11 11

Question 2

```
#include <stdio.h>
int num=10;
void update(int *ptr)
{
   *ptr+=num;
}
int main(void)
{
   int num=100;
   printf("%d\n",num);
   update(&num);
   printf("%d\n",num);
}
```

- A) 100 100
- B) 100 10
- C) 100 110
- D) 100 garbage

Question 3

```
#include <stdio.h>
int main(void)
{
   int arr[5]={10,20,30,40,50};
   int *ptr;
   ptr = arr+2;
   *ptr=33;
   *ptr++;
   printf("%d",--*ptr);
}
```

- A) 31
- B) 20
- C) 39
- D) 40

Question 4

```
#include <stdio.h>
int main(void)
{
   char num=256;
   int *ptr=&num;
   *ptr++;
   int *ptr2=--ptr;
   printf("%d",*(char *)ptr2);
}
```

- A) Garbage
- B) 10
- C) 0
- D) 11

Question 5

(Assume x address is 2000 and ptr address is 2004)

```
#include <stdio.h>
int main()
{
   int x = 0 ;
   int *ptr = &x;
   *ptr += 5;
   printf("\nx = %d", x);
   printf(" *ptr = %d", *ptr);
   (*ptr)++;
   printf("\nx = %d", x);
   printf(" *ptr = %d", *ptr++);
   printf("\n Difference= %d", ptr - &x);
   return 0;
}
```

- A) x=5 *ptr=5 x=6 *ptr=6 Difference=1
- B) x=5 *ptr=garbage x=6 *ptr=garbage Difference=2
- C) x=5 *ptr=garbage x=6 *ptr=6 Difference=-4
- D) x=5 *ptr=5 x=6 *ptr=6 Difference=4

Question 6

```
#include <stdio.h>
int main()
{
   int arr[10];
   int *i = &arr[2], *j = &arr[5];
   int diff = j - i;
   printf("%d", diff);
}
```

- A) 3
- B) 6
- C) Garbage Value
- D) Error

Question 7

```
#include <stdio.h>
int a=3;
void n(int x)
{
   x = x*a;
   printf("%d ",x);
}
void m(int y)
{
   a =1;
   a =y-a;
   n(a);
   printf("%d ",a);
}
int main()
{
   m(a);
}
```

- A) 6, 2
- B) 6, 6
- C) 4, 2
- D) 4, 4

Question 8

```
#include <stdio.h>
void flo(int *p, int *q)
{
   p = q;
   *p = 2;
}
int l =0,j = 1;
int main()
{
   flo(&l,&j);
   printf("%d %d \n",l,j);
   return 0;
}
```

- A) 2 2
- B) 2 1
- C) 0 1
- D) 0 2

Question 9

```
#include <stdio.h>
void main()
{
   int a = 5 ;
   int *b = &a;
   int **c = &b ;
   printf("%d %d %d", a, ++*b, ++**c);
}
```

- A) 7 7 7
- B) 5 6 6
- C) 7 7 6
- D) 7 6 5

Question 10

```
#include <stdio.h>
int main()
{
   const int a =7;
   const int * p=&a;
   printf("%d",++*p);
   return 0;
}
```

- A) 7
- B) 8
- C) Compiler Error
- D) Garbage Value8.

CPROGRAM QUIZ – 7

Question 1 What is the output of the C program with #define?

```
#define Q 156
int main()
{
   int a=10;
   a = a*'Q';
   printf("a=%d",a);
   return 0;
}
```

- 1560
- 156
- 810
- 1130

Question 2 What is the output of the C program?

```
#include<stdio.h>
#define CONDITION(x)\
printf("Machine Learning\n"+8);
int main()
{
   CONDITION(0);
   return 0;
}
```

- Runtime Error
- Compilation error
- Learning
- 0

Question 3 What is the return type of malloc() or calloc()?

- int *
- int **
- void *
- void **

Question 4 Can you combine the following two statements into one? char *p; p = (char*) malloc(100);

- char p = *malloc(100);
- char *p = (char) malloc(100);
- char *p = (char*)malloc(100);
- char *p = (char *)(malloc*)(100);

Question 5 What is a dangling pointer in C?

- if pointer is pointing to a memory location to which memory already deallocated
- if pointer is pointing to more than one memory location
- if pointer is not defined properly
- none of above

Question 6 What is the output of this code?

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
   char *ptr=NULL;
   int alloc_length =16 ;
   int new_length;

   ptr = (char *)malloc(sizeof(char)*alloc_length);

   ptr = realloc(ptr, alloc_length << 1);

   free(ptr);
   ptr=NULL;

   return 0;
}
```

- allocated 16bytes and free 32bytes
- allocated 32bytes and free 32bytes
- allocated 32bytes and free 16bytes
- allocated 16bytes and free 16bytes

Question 7 What is the output of this code?

```
#include<stdio.h>
#include<string.h>
#include<stdlib.h>

int main()
{
   char *s;
   char *fun();
   s = fun();
   printf("%s\n", s);
   return 0;
}
char *fun()
{
   char *buffer;
   buffer = (char*)malloc(sizeof(char)*5);
   strcpy(buffer, "RAM");
   return (buffer);
}
```

- Segmentation fault
- compile time error :function returns address of local variable
- RAM
- none of these

Question 8 What will be the output of this code?

```
#include<stdio.h>
#define a 100
int main()
{
   printf("a : %d\n",a);
   #ifndef a
   #define a 30
   #else
   #undef a
   #define a 40
   printf("a : %d",a);
   return 0;
}
```

- a : 100 a : 30
- a : 0 a : 100
- a : 100 a : 40
- Compile time error

Question 9 What will be the output of this code?

```
#include<stdio.h>
#define a 10
#define z(a) a-1
#define SQR(x) (x*x)
int main()
{
   int c, b=3;
   c = z(a) * SQR(b+2);
   printf("%d\n", c);
   return 0;
}
```

- -1
- 225
- 99
- 5

Question 10 Dynamic memory allocation is done on:

- stack section
- Heap section
- Data section
- Register

CPROGRAM QUIZ – 8

Question 1

```
#include <stdio.h>
int main()
{
   union values
   {
      unsigned char a;
      unsigned char b;
      unsigned int c;
   };
   union values val;
   val.a=0;
   val.b=0;
   val.c=1;
   printf("%d,%d,%d",val.a,val.b,val.c);
   return 0;
}
```

- 0,0,1
- Error
- 1,1,1
- 1,0,0

Question 2

```
#include <stdio.h>
int main()
{
   union test
   {
      int i;
      int j;
   };
   union test var=10;
   printf("%d,%d\n",var.i,var.j);
}
```

- 10,10
- 10,0
- 0,10
- Error

Question 3 What is the output of this C code?

```
#include <stdio.h>
struct p
{
   int k;
   char c;
   float f;
};
int main()
{
   struct p x = {.c = 166, .f = 3, .k = 1};
   printf("%f\n", x.f);
}
```

- 3.000000
- Compile time error
- Undefined behavior
- 1.000000

Question 4 What will be the output of following program?

```
#include <stdio.h>
struct sample
{
   int a=1;
   char b='S';
   float c=11.5;
};
int main()
{
   struct sample s;
   printf("%d,%c,%f",s.a,s.b,s.c);
   return 0;
}
```

- Error
- 1,S,11.5
- 1,S,11.500000
- No Error, No Output

Question 5 What will be the output of following program?

```
#include <stdio.h>
struct student
{
   int no;
   char name[10];
}
main()
{
   struct student s;
   s.no = 12;
   printf("hello");
   return s;
}
```

- Compile time error
- Nothing
- hello
- exit value 12

Question 6 The correct syntax to access the member of the ith structure in the array of structures is?

```
struct temp
{
   int b;
}s[50];
```

- s.b[i];
- s.[i].b;
- s.b[i];
- s[i].b;

Question 7 What is the correct syntax to declare bit-field in structure?

- struct temp { unsigned int a : 1; }s;
- struct temp { unsigned int a = 1; }s;
- struct temp { unsigned float a : 1; }s;
- None of the mentioned

Question 8 Which of the following is not true in case of structures?

- structure is a user defined data type.
- size of a structure is sum of size of each of its members.
- We can declare a structure within a function.
- We can initialize the members of structure at the time of structure declaration.

Question 9

```
#include<stdio.h>
struct employee
{
   int emp_id;
   char name[20];
   double salary;
};
int main()
{
   struct employee emp;
   struct employee *ptr = &emp;
   printf("Enter the details :");
   scanf("%d%s%lf",&emp.emp_id,emp.name,&emp.salary);
   return 0;
}
```

Which of the following is the correct way to access the members of the structure (eg: name) using a pointer ptr?

- ptr.name
- ptr->name
- &ptr->name
- *ptr->name

Question 10 What will be the output of the code below?

```
#include<stdio.h>
struct student
{
   int roll_no;
   char name[20];
   int marks;
};
int main()
{
   struct student s1[5];
   printf("size of structure :%d" ,sizeof(s1));
   return 0;
}
```

- 28
- 9
- 140
- 45

CPROGRAM QUIZ – 9

1. What is the output of the following program?

```
int main()
{
   char c;
   FILE *fp;
   fp=fopen("sunbeam.txt","r");
   while((c=fgetc(fp))!=EOF)
      printf("%c",c);
   fclose(fp);
   return 0;
}
```

- It will print the content of file sunbeam.txt
- It will print the content of file till it encounter new line character
- Compilation Error
- None of the above

2. What is the operator used to make 1's complement?

- & Bitwise AND Operator
- | Bitwise OR operator
- ~ Bitwise Negate Operator
- ^ Bitwise Exclusive OR

3. What is the output of the following code?

```
int main(){
   int a,b,c;
   a=4;
   b=2;
   c=a<<b && a>>b;
   printf("%d",c);
}
```

- 1
- 0
- 4
- 2

4. What is the output of the following code?

```
int main()
{
   int a,b,c;
   a=4;
   b=2;
   c=~(a^b / a<<b);
   printf("%d",c);
}
```

- 5
- 10
- -5
- -10

5. What will be the output of the following code and text file data? File (test_file.txt): C is a general-purpose, 0 procedural computer programming language supporting structured programming 0, lexical variable scope, and recursion, with a static type system 0.

```
#include <stdio.h>
int main(){
   char c;
   FILE *fp;
   fp=fopen("test_file.txt","r");
   while((c=fgetc(fp))!= 0 )
      printf("%c",c);
   fclose(fp);
   return 0;
}
```

- Print the contents of file "test_file.txt"
- Print the contents of file "test_file.txt" upto '0' character in "test_file.text"
- Infinite loop
- Error in the code.

6. What will be the output of the following code and text file? File (t2.txt):
ABCDEFGHIJKLMNOP

```
#include <stdio.h>
int main( void )
{
   FILE *fp=NULL;
   char buf[10];
   fp = fopen("t2.txt", "r");
   fseek(fp, +4L, SEEK_CUR);
   fgets(buf,5,fp);
   fputs(buf,stdout);
   return 0;
}
```

- ABCDE
- ABCD
- EFGH
- EFGHI

7. If we want to add a record in an existing file at the end of the file, what is the preferred mode?

- r
- w
- r+
- a

8. What will be the output of the following code?

```
#include <stdio.h>
int main ()
{
   FILE *fp;
   int c;
   fp = fopen("file.txt","r");
   while(1)
   {
      c = fgetc(fp);
      if( feof(fp) )
      {
         break ;
      }
      printf("%c", c);
   }
   fclose(fp);
   return(0);
}
```

- it will print all the contents of file.
- doesn't print anything.
- Infinite loop
- compiler error

9. Which of the following is the correct syntax for fscanf and fprintf?

- fprintf("format specifier",variables,fp); fscanf("format specifier",variables,fp);
- fprintf(fp,count,"format specifier",variables); fscanf(fp,count,"format specifier",variables);
- fprintf(fp,"format specifier",variables); fscanf(fp,"format specifier",variables);
- None of the above.

CPROGRAM QUIZ – 10

Question 1: What will be the output of the following program?

```
#include <stdio.h>

int main()
{
   int num1 = 0;
   int num2 = 2;
   int result;

   result = ++num1 && num2-- && --num1 || ++num2;

   printf("%d %d %d", num1, num2, result);

   return 0;
}
```

Options:

- A) 1 2 0
- B) 0 2 1
- C) 1 2 1
- D) 0 2 2

Question 1: What will be the o/p of the following program?

```
#include<stdio.h>
#include<string.h>
int main()
{
   int a=11;
   if(a>10)
      printf("Value of a is %d ",a);
      printf("Value is greater than 10");
   else
      printf("Value is less than 10");
      printf("Outside the if statement");
}
```

- Value of a is 10 Value is greater than 10
- Value of a is 10 Outside the if statement
- Compile time Error
- Value is greater than 10 Outside the if statement

Question 2: What will be the o/p of the following code?

```
#include<stdio.h>
#include<string.h>
int main()
{
   int result=0;
   result = printf(" ");
   switch (result)
   {
      case 0:
         printf("Hello 0 ");
         break;
      case 1:
         printf("Hello 1 ");
         break;
      default:
         printf("Default case");
         break;
   }
}
```

- Hello 0
- Default case
- No Output
- Hello 1

Question 3: What will be the o/p of the following code?

```
#include<stdio.h>
int main()
{
   int result;
   char name[10]="Sun\0beam";
   result = strlen(name);
   printf("%d",result++);
}
```

- 4
- 3
- 8
- 9

Question 4: What will be the o/p of the following code?

```
#include<stdio.h>
int main()
{
   int num=10;
   int * const ptr = NULL;
   ptr=&num;
   printf("%d",*ptr);
   printf("%d",num);
}
```

- 10 10
- 10 0
- Compile time Error
- 0 0

Question 5: What is constant in the following snippet? int * const ptr;

- Value is constant
- Both value and pointer is constant
- Pointer is constant
- No such declaration allowed

Question 6: What will be the o/p of the following code?

```
#include <stdio.h>
#include<stdlib.h>
int main(void)
{
   char src[10]="Sunbeam";
   char dest[10];
   strcpy(dest,src+1);
   strcat(dest,strcpy(dest,dest+2));
   strcpy(src,dest);
   printf("%s",src);
   return 0;
}
```

- Sunbeam
- beam
- nbeam
- beambeam

Question 7: What will be the o/p of the following code?

```
#include<stdio.h>
void show(void);
int main(void)
{
   int i;
   for (i = 0;i<=5; ++i)
   {
      if(i==5)
         goto display;
      printf("%d",i++);
   }
}
void show(void)
{
   display:
   printf("Hello");
}
```

- 2 4 6 Hello
- 0 2 5 Hello
- Compile time Error
- 0 2 4 6

Question 8: What will be the o/p of the following code?

```
#include<stdio.h>
struct bit
{
   unsigned int a:3;
   int b:3;
};
int main()
{
   struct bit b1;
   b1.a=10;
   b1.b=5;
   printf("%d",b1.a);
   printf("%d",b1.b);
}
```

- 1 -2
- 2 -3
- 1 -2
- 1 1

Question 9: What will be the o/p of the following code?

```
#include<stdio.h>
int show(int num1 , int num2);
static int result;
int main()
{
   int ans;
   int num1 = 1;
   int num2 = 2;
   ans = show(num1,num2);
   printf("%d %d",ans,result++);
}
int show(int num1 , int num2)
{
   int result;
   result = num1 + num2;
   return num2++,result;
}
```

- 3 3
- 3 0
- 2 0
- 3 1

Question 10: What will be the o/p of the following code?

```
#include<stdio.h>
#define EXECUTE(x,y) x##y
int main()
{
   int a = 200,b=300,ab=400;
   printf("%d",ab+EXECUTE(a,b));
   return 0;
}
```

- 1000
- 800
- 900
- Compile time Error

Question 11: What will be the o/p of the following code?

```
#include<stdio.h>
int main()
{
   int i;
   for (i = 0; i < 3; )
   {
      i++;
   }
   if(i==3)
   {
      break;
      printf("Hello");
   }
   printf("World");
   return 0;
}
```

- World
- HelloWorld
- Compile time Error
- Runtime Error

Question 12: What will be the o/p of the following code?

```
#include<stdio.h>
#include<string.h>
int main()
{
   char *str[] = {"DAC", "DMC", "DBDA", "DESD", "DITTIS", "PRECAT"};
   printf("%d, %d, %d", sizeof(str), strlen(str[5]),sizeof(str[5]));
   return 0;
}
```

- 24, 6, 4
- 24, 7, 7
- 24, 6, 7
- 4, 6, 4

Question 13: What will be the o/p of the following code?

```
#include <stdio.h>
enum Season { SPRING = 1, SUMMER, AUTUMN = 4, WINTER };
int main()
{
   enum Season currentSeason = SUMMER;
   switch (currentSeason , currentSeason + 1) {
      break;
      case SPRING:
         printf("It's Spring!");
         break;
      case SUMMER:
         printf("It's Summer!");
      case AUTUMN:
         printf("It's Autumn!");
         break;
      case WINTER:
         printf("It's Winter!");
         break;
      default:
         printf("Indian season.");
   }
   return 0;
}
```

- It's Spring
- It's Summer
- Error due to break without case
- Indian season

Question 14: What will be the o/p of the following code?

```
#include<stdio.h>
int main()
{
   if(~printf("sunbeam "))
      printf(" Pune");
   else
      printf("Karad");
}
```

- sunbeam
- karad
- pune
- sunbeam pune

Question 15: What will be the o/p of the following code?

```
#include<stdio.h>
int main()
{
   #ifndef max
   printf("sun");
   printf("beam");
}
```

- beam
- sun
- sunbeam
- complier error

Question 16: What will be the output of the following code?

```
#include <stdio.h>
typedef struct {
   int x;
   int y;
} Point;
void modify(Point p) {
   p.x = 10;
   p.y = 20;
}
int main() {
   Point p = {1, 2};
   modify(p);
   printf("%d %d\n", p.x, p.y);
   return 0;
}
```

- 10 20
- 1 2
- Garbage value Garbage value
- Compilation error

Question 17: What will be the output of the following code?

```
#include <stdio.h>
int main() {
   int arr[2][3] = {{1, 2, 3}, {4, 5, 6}};
   int* ptr = &arr[0][1];
   printf("%d\n", ptr[4]);
   return 0;
}
```

- 4
- 5
- 6
- Garbage value

Question 18: What is the purpose of the strchr() function in C?

- To concatenate two strings
- To compare two strings
- To find the length of a string
- To find the first occurrence of a character in a string

Question 19: What will be the output of the following recursive code?

```
#include <stdio.h>
void print(int n) {
   if (n > 0) {
      print(--n);
      printf("%d ", n);
   }
}
int main() {
   print(5);
   return 0;
}
```

- 1 2 3 4 5
- 0 1 2 3 4
- 5 4 3 2 1
- -1 0 1 2 3

DS QUIZ – 1

Question 1: State whether the following statement is True or False:

"Linear search is rarely used practically because other search algorithms, such as the binary search algorithm, have significantly faster searching comparisons."

- True
- False

Question 2: Consider the array arr[15] = {1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15}. Suppose we are searching for the key 1 using binary search. After how many comparisons will the key be found?

- 1
- 2
- 3
- 4

Question 3: What is the exit condition of a binary search when start is the beginning index and end is the ending index of the sorted array?

- start != end
- end == start
- end > start
- end < start

Question 4: Which searching algorithm generally uses a lower number of LOC (Lines of Code)?

- Sequential Search
- Simple Search
- Binary Search
- None of these

Question 5: Which of the following is the tightest upper bound representing the number of swaps required to sort n numbers using Selection Sort?

- O(log n)
- O(n log n)
- O(n)
- $O(n^2)$

Question 6: An array of 35 distinct elements is to be sorted using Quick Sort. Assume the pivot is chosen uniformly. What is the probability that the pivot element gets placed in the worst possible location in the first round of partitioning (rounded to 2 decimal places)?

- n/2
- (n-1)/2
- (n+1)/2
- None of these

Question 7: Sort the following list in ascending order using Bubble Sort: {12, 17, 33, 13, 24, 10}. What will be the sequence after the 2nd pass?

- 12 13 17 10 24 33
- 12 13 17 24 33 10
- 12 17 13 10 24 33
- 10 12 13 17 24 33

Question 8: The Quick Sort algorithm requires a maximum of _____ stack space.

- O(n)
- O(n^2)
- O(log n)
- None of these

Question 9: Which algorithm can be visualized as people in a queue wanting to stand in a height-wise sorted manner, swapping positions among themselves until everyone is in increasing order of height?

- Bubble Sort
- Selection Sort
- Heap Sort
- Merge Sort

Question 10: The following formula describes which data structure?

left_subtree key < root key <= right_subtree keys

- Binary Tree
- AVL Tree
- Binary Search Tree
- None of the above

DS QUIZ – 2

Question 1: Time Complexity of search operation on priority queue is:

- O(1)
- O(n-1)
- O(n)
- O(log n)

Question 2: Let Q denote a queue containing 4 numbers i.e. [11, 22, 33, 44] and S be an empty stack. head(Q) returns the element at the head of the queue Q without removing it from Q. Similarly top(S) returns the elements at the top of S without removing it from S. Consider the algorithm below:

```
if S is empty or top(S) <= head(Q) then
   x = dequeue(Q)
   push(S,x)
else
   x = pop(S)
   enqueue(Q,x)
end
```

Which element will be at the front of the queue after execution of this code two times?

- 11
- 22
- 33
- 44

Question 3: What is the time complexity to delete an element from the front end of the queue?

- O(n)
- O(1)
- O(log n)
- O(n log n)

Question 4: The initial value of top in a stack is maintained as:

- -1
- 0
- 1
- garbage

Question 5: Array implementation of Stack is not dynamic, which of the following statements supports this argument?

- Space allocation for array is fixed and cannot be changed during run-time
- User unable to give the input for stack operations
- A runtime exception halts execution
- Improper program compilation

Question 6: Traffic system uses ------

- Linear Queue
- Circular Queue
- Priority Queue
- DEQueue

Question 7: A queue is implemented using a linear singly linked list. The queue has a head pointer and a tail pointer. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail. Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue operations respectively, for this data structure?

- θ(1), θ (1)
- θ (1), θ (n)
- θ (n), θ (1)
- θ (n), θ (n)

Question 8: Find Infix Expression for the given postfix expression:

A B C D * - E / + F + G H / -

- A / ((B - C * D) + E) + F - G / H
- A + ((B - C * D) / E) + F - G / H
- A + ((B - C * E) / D) + F - G / H
- A + ((B - C * D) / E) + F / G - H

Question 9: What does stack underflow mean?

- Accessing items from undefined stack.
- Adding items to a full stack.
- Deleting items from empty stack.
- Index out of bound exception.

Question 10: Circular queue is also known as:

- Standard Buffer
- Cyclic Buffer
- Ring Buffer
- Radius Buffer

DS QUIZ – 3

Question 1 Assume that a structure for a Binary Search Tree exists. What does the following function do?

```
int function(root)
{
  current = root;
  while (current->left != NULL)
  {
    current = current->left;
  }
  return(current->data);
}
```

- Left most child of BST.
- Right most child of BST.
- subtrees of BST
- None above mentioned

Question 2 Which of the following is not application of graph?

- Coloring of maps algorithms
- Job scheduling
- Web Page navigation
- Redo / Undo actions

Question 3 Which of the following is not a property of a binary search tree?

- The left subtree of a node contains only smaller values.
- The right subtree of a node contains only larger values.
- The values in the left and right subtrees are equal.
- All nodes in the left subtree are less than the node, and all nodes in the right subtree are greater.

Question 4 What is true about binary tree?

- A binary tree is a type of tree in which every node has either no child node or one child node or two child nodes.
- A binary tree is an important class of a tree data structure in which a node can have at most two children.
- Both A and B
- None of above

Question 5 Find out Binary Search Tree by inserting following numbers from Left to Right? **11, 6, 8, 19, 4, 10, 5, 17, 43, 49, 31**

- image option 1
- image option 2
- image option 3
- None Of Above

Question 6 We can say that a subtree is left-heavy if the balance factor:

- Is greater than zero
- Is less than zero
- Equals to zero
- None of these.

Question 7 A binary tree whose every node has either zero or two children is called:

- Complete binary tree
- Binary search tree
- Extended binary tree
- None of above

Question 8 How to represent Adjacency Matrix in graph?

- It is Matrix a[v][v] where v is no of vertices. a[i][j] = 1 if i and j are adjacent = 0 otherwise
- It is Matrix a[v][v] where v is no of edges. a[i][j] = 1 if i and j are not adjacent = 0 otherwise
- Both A and B are correct
- None of ABOVE

Question 9 Which of the following is true?

- A graph may contain no edges and many vertices
- A graph may contain many edges and no vertices
- A graph may contain no edges and no vertices
- A graph may contain no vertices and many edges

Question 10 A graph with all vertices having equal degree is known as a _____.

- Multi Graph
- Regular Graph
- Simple Graph
- Complete Graph

DS QUIZ – 4

Question 1 What is the time complexity of searching for an element in a circular linked list?

- O(n)
- O(1)
- O(n log n)
- O(n log (n-1))

Question 2 Breadth First Search Traversal follows which functionality?

- LIFO
- FIFO
- Pre-Order
- Recursive

Question 3 An algorithm is _____

- A stepwise solution to a problem
- A problem
- A real life mathematical problem
- A pictorial representation of stepwise solution to problem

Question 4 A _____ node is a special node that is found at the beginning of the linked list.

- main
- header
- start
- beginner

Question 5 To stop traversing a doubly circular linked list from head, which condition is correct?

- ptr != NULL
- ptr->next != head
- ptr != head
- ptr->next != null

Question 6 The no of external nodes in a full binary tree with n internal nodes is?

- n
- n+1
- 2n
- 2n+1

Question 7 What is the worst case time complexity of inserting n elements into an empty linked list, if the linked list needs to be maintained in sorted order?

- O(n)
- $O(n^2)$
- O(log n)
- O(1)

Question 8 A linear list of elements in which deletion can be done from one end (front) and insertion can take place only at the other end (rear) is known as a _____

- queue
- stack
- tree
- linked list

Question 9 If multiple values are competing for the same slot in a hashtable, what is it called?

- Duplication
- Replication
- Collision
- Diffusion

Question 10 Which sorting algorithm's time complexity remains $O(n^2)$ in its best, average, and worst cases?

- Selection sort
- Merge sort
- Bubble sort
- Insertion sort

Question 11 If a doubly linked list contains nodes 112, 221, 454, 455, 554, 545, 121 and a pointer p points to node 554, then what would p -> prev -> next -> next -> prev fetch?

- 121
- 545
- 554
- 454

Question 12 What are the number of nodes of left and right subtree of the binary tree if the data is inserted in the following order: 45, 15, 8, 5, 6, 5, 65, 47, 12, 18, 10, 73, 50, 16, 61

- 7 6
- 6 7
- 8 5
- 5 8

Question 13 When two different keys map to the same index in a hash table, what is this phenomenon called?

- Clustering
- Collision
- Diffusion
- Confusion

Question 14 The total number of comparisons in a bubble sort is

- O(n log n)
- O(n^2)
- O(2n)
- None of the above

Question 15 Worst case time complexity of Binary Search is _____

- O(n log n)
- O(log n)
- O(n log 1)
- O(n*n)

Question 16 In a circular linked list:

- Components are all linked together in some sequential manner.
- There is no beginning and no end.
- Components are arranged hierarchically.
- Forward and backward traversal within the list is permitted.

Question 17 Which of the following is useful when data has to be stored and retrieved in reverse order?

- Stack
- Queue
- Linked List
- Tree

Question 18 Which analysis refers to computing the running time of any operation in mathematical units of computation?

- Symptotic
- Asymptotic
- Regular
- Notational

Question 19 If the in-order traversal of a binary tree is T, Q, U, P, V, R, W and its post-order traversal is T, U, Q, V, W, R, P. What would be the pre-order traversal of this tree?

- P, Q, T, U, R, W, V
- P, Q, T, R, V, U, W
- P, T, Q, U, R, V, W
- P, Q, T, U, R, V, W

Question 20 Two main measures for the efficiency of an algorithm are:

- Time and space
- Processor and memory
- Complexity and capacity
- Data and space

CPP QUIZ – 1

Question 1: Inline Functions Which of the following statement is correct in context of inline function?

1. Inline functions gets process during compile time
2. If we implement member function inside class then by default it is considered as inline
3. If function is inline we cannot separate its code in multiple files.
4. If we want to reduce overhead to the compiler during function call then we should declare function as a inline

- Only 1
- Both 1 and 2 are correct
- Both 1 and 3 are correct
- All 1,2,3 and 4 are correct

Question 2: Class Output What is the output of this program?

```
class date
{
   int day,month,year;
   date()
   {
      day=25;
      month=4;
      year=2020;
   }
public:
   void printDate()
   {
      cout<<this->day<<"-"<<this->month<<"-"<<this->year<<endl;
   }
};

Int main()
{
   date today;
   today.printDate();
   return 0;
}
```

- 25-4-2020
- Garbage Values
- Compilation Error
- Runtime Error

Question 3: Which of the following statement is correct in context of object?

1. Value stored inside object is called state of the object
2. Property of an object which distinguishes it from all other objects is called state of the object
3. Member function of class represent behaviour of object

- Only 1
- Both 1 and 2 are correct
- Both 1 and 3 are correct
- All of the above

Question 4: Which of following statements is true in context of function overloading?

- Function defined with same name and different signature within same scope called as function overloading
- Except main() function and destructor we can overload all the functions
- Function overloading concept implicitly uses mangled name.
- Return type is not considered in function overloading.
- All of the above

Question 5: _____ are member functions of the class, which is used to read the state of the object.

- constructor
- Mutators
- Iterator
- Inspectors
- Destructor

Question 6: Which of the following permits function overloading in c++?

- Type of arguments
- Number of arguments
- Type, Number and Sequence of arguments
- Return type of function
- Both C and D

Question 7: Which of the following statements is/are correct in context of class?

1. Class is a collection of data member and member functions.
2. Class is a physical entity.

- Only 2 is correct
- Only 1 is correct
- Both 1 and 2 are correct
- None of the above

Question 8: In C++, by default structure members are ____ and class members are _____.

- private, private
- private, public
- public, public
- public, private

Question 9: Which of the following is a properly defined structure?

- struct {int t;}
- struct a_struct {int a;};
- struct p_struct int p;
- struct s_struct {int s;}

Question 10: _____ operator is designed to use with cin.

- <
- <<
- >>
- >

Question 11: What will be the o/p of following code:

```
#include <stdio.h>
int main(void)
{
   int arr[3][2]={{1,2},{1},{5}};
   int i,j;
   int *ptr=*(arr+2);
   printf("\n %d",*++ptr+1);
}
```

- 1
- 0
- Garbage
- Compiletime error

Question 12: Array Declaration What is the result of the following code:

```
#include<stdio.h>
#define COL 3
#define ROW 3
int main(void)
{
   int arr[ROW][] = {{10,20,30},{11,22},{111}},i,j;
   for(i=0; i<3; i++)
   for(j=0; j<3; j++)
   printf("%d",arr[i][j]);
   return 0;
}
```

- Compile time error
- Run time error
- Prints Array Elements (1020301122011100)
- Prints array elements (1020301122111)

Question 13: What is the output of this code:

```
#include<stdio.h>
int main( void )
{
   int arr[2][3] = {10,20,30,40};
   int *ptr[] = {(int *)arr+2, (int *)arr+1};
   int **p=ptr;
   printf("%d %d", p[0][1], *(*(p + 1) + 0));
   return 0;
}
```

- 40 20
- 40 40
- 30 20
- 20 20

Question 14: If i/p is given as <programname> monday tuesday wednesday, what will be the o/p?

```
#include<stdio.h>
int main( int argc,char* argv[])
{
   int i=1;
   printf("%c",++*argv[i]);
   return 0;
}
```

- o
- n
- t
- u

CPP QUIZ – 2

Question 1 Which of the following statements is correct?

1. A reference is useful, when we intend to change the values of actual arguments through the function call.
2. A reference is useful, when we want to save memory by preventing the creation of large structure variable that are being passed to the function.

- Only 1 is correct.
- Only 2 is correct.
- Both 1 and 2 are correct.
- Both 1 and 2 are incorrect.

Question 2 Pick the correct statement about references.

- References can be assigned value NULL
- References once initialized, we cannot change its referent variable
- Reference should not be initialized when created
- Reference is the same as pointers

Question 3 Which of the following is used to release the dynamically allocated memory in CPP if we allocate memory using operator?

- remove
- free()
- delete
- Both 2 and 3

Question 4 Which of the following statements are correct in context of below code?

```
#include<iostream>
using namespace std;
int main( void )
{
   int *ptr = new int(3);
   return 0;
}
```

- we are allocating memory for a single variable but memory will be initialized with value 3.
- we are allocating memory for array of three integer variable
- we are allocating memory for array of three integer variable but memory will be initialized with 0 value
- None of the above

Question 5 Which of the following is/are valid ways to allocate memory for an integer by dynamic memory allocation in CPP?

- int *p = new int(100);
- int *p; p = new int; *p = 100;
- int *p = NULL; p = new int; *p = 100;
- Only 1 and 2
- All of these

Question 6 In which situations a destructor is called?

- we allocate memory using the new operator and free it using delete operator.
- object goes out of scope
- program terminated
- both A and B

Question 7 When a function is defined inside a class, this function is called _____.

- Inside function
- Member function
- Inline function
- Outline function

Question 8 A destructor takes _____ arguments.

- 0
- 1
- 2
- 3

Question 9 A variable can be declared as reference by putting _____ in the declaration.

- #
- $
- &
- *

Question 10 If an object is passed by reference, the changes made in the function _____.

- Are reflected to the main object of caller function too
- Are reflected only in local scope of the called function
- Are reflected to the copy of the object that is made during pass
- Are reflected to caller function object and called function object also

CPP QUIZ – 3

Question 1: What is the index number of the last element of an array with 9 elements?

- 9
- 8
- 0
- 1

Question 2: Which of the following gives the address of the first element in array?

- array[0];
- array[1];
- array(2);
- array;

Question 3: If a copy constructor is not defined in a class then:

- Compiler itself defines one
- Compiler gives Syntax Error
- Compiler gives Run time Error
- None of Above

Question 4: Syntax for Copy constructor is:

- classname (const classname &obj) { // body of constructor }
- classname() { // body of constructor }
- Both A and B
- None of above

Question 5: _____ is called when a new object is created from an existing object, as a copy of the existing object.

- Copy Constructor
- Default Constructor
- Assignment operator
- None of above

Question 6: Default copy constructor does _____:

- Deep Copy
- Shallow copy
- Both A and B
- None of Above

Question 7: Which of the following scenario deep copies generally required?

- A constructor to either make an initial allocation or set the pointer to NULL.
- A destructor to delete the dynamically allocated memory.
- A copy constructor to make a copy of the dynamically allocated memory.
- An overloaded assignment operator to make a copy of the dynamically allocated memory.
- All of above

Question 8: When a copy constructor may be called?

- When an object of the class is returned by value.
- When an object of the class is passed (to a function) by value as an argument.
- When an already created object is assigned to newly created object of the same class
- When the compiler generates a temporary object.
- All of the above

Question 9: Copy constructor is a member function of a class which gets called implicitly:

- when we pass an object to the function by value.
- when we return an object from function by value.
- when we assign an already created object to the newly created object.
- All of the above

Question 10: What is shallow copy?

- A shallow copy creates a copy of the dynamically allocated objects too.
- A shallow copy just copies the values of the data as they are.
- A shallow copy creates a copy of the statically allocated objects too
- A shallow copy is used to rename object.

CPP QUIZ – 4

Question 1 An operator function is created using _____ keyword.

- iterator
- allocator
- constructor
- operator

Question 2 Which of the below mentioned can be made constant?

- Member functions of a class
- Function arguments
- Class data member
- All of Above

Question 3 The static member functions _____.

- Can be called using class name
- Can be called using program name
- Can be called directly
- Can't be called outside the function

Question 4 Syntax for making member function as Constant is _____.

- void display()
- void display() const
- const void display()
- void const display()

Question 5 When will we use the function overloading?

- same function name but different number of arguments
- different function name but same number of arguments
- same function name but same number of arguments
- different function name but different number of arguments

Question 6 Which keyword must be used to declare a member function as a constant member function?

- constant
- const
- functionConst
- unchanged

Question 7 Which of the following is true?

- Static methods cannot be overloaded.
- Static data members can only be accessed by static methods.
- Non-static data members can be accessed by static methods.
- Static member function can only access static members (data and methods)

Question 8 Which of the following statements is/are correct in the context of constant data members?

1. If we declare a data member constant then it is mandatory to initialize it using constructors member initializer list.
2. If we don't want to modify state of the data member inside any member function of the class including constructor body then we should declare data member constant.

- Only 1 is correct
- Only 2 is correct
- Both 1 and 2 are correct
- None of the above

Question 9 Which of the following operator(s) cannot be overloaded?

- . (Member Access or Dot operator)
- ?: (Ternary or Conditional Operator)
- :: (Scope Resolution Operator)
- All of the above

Question 10 when we declare data member as constant it is compulsory to initialise that data member inside _____.

- constructor initializer list
- constructor(using assignment operator inside constructor code)
- member function
- global function

CPP QUIZ – 5

Question 1 Which of the following support the concept of hierarchical classification?

- Polymorphism
- Encapsulation
- Inheritance
- Abstraction

Question 2 Which feature of OOPS illustrated the code reusability?

- Polymorphism
- Abstraction
- Encapsulation
- Inheritance

Question 3 Which inheritance type is used in the class given below? class A: public X, public Y {}

- Multilevel inheritance
- Multiple inheritance
- Single Inheritance
- None of above

Question 4 Function Overloading comes under?

- Compile Time Polymorphism
- Runtime Polymorphism
- Abstraction
- None of the Above

Question 5 Which of the following is minor pillars of OOP?

- Typing
- Persistance
- Concurrency
- All of the above

Question 6 If there is a "is-a" relationship exist between two types we use?

- Aggregation
- Inheritance
- Composition
- None of the above

Question 7 Default mode of inheritance is

- Private
- Public
- Protected
- None

Question 8 If single base class is having multiple derived classes then such inheritance is called?

- Multiple Inheritance
- Hierarchical Inheritance
- Single Inheritance
- Multilevel Inheritance

Question 9 When the call to the function is resolved at run time it is called run time polymorphism, and it is achieved by using _____.

- function overriding
- function overloading
- operator overloading
- function overloading and operator overloading

Question 10 What is true about the protected mode of inheritance?

- public member of base becomes protected members of derived class
- protected members of base become protected members of derived class
- private members of base become private members of derived class [not accessible in derived]
- All of the above